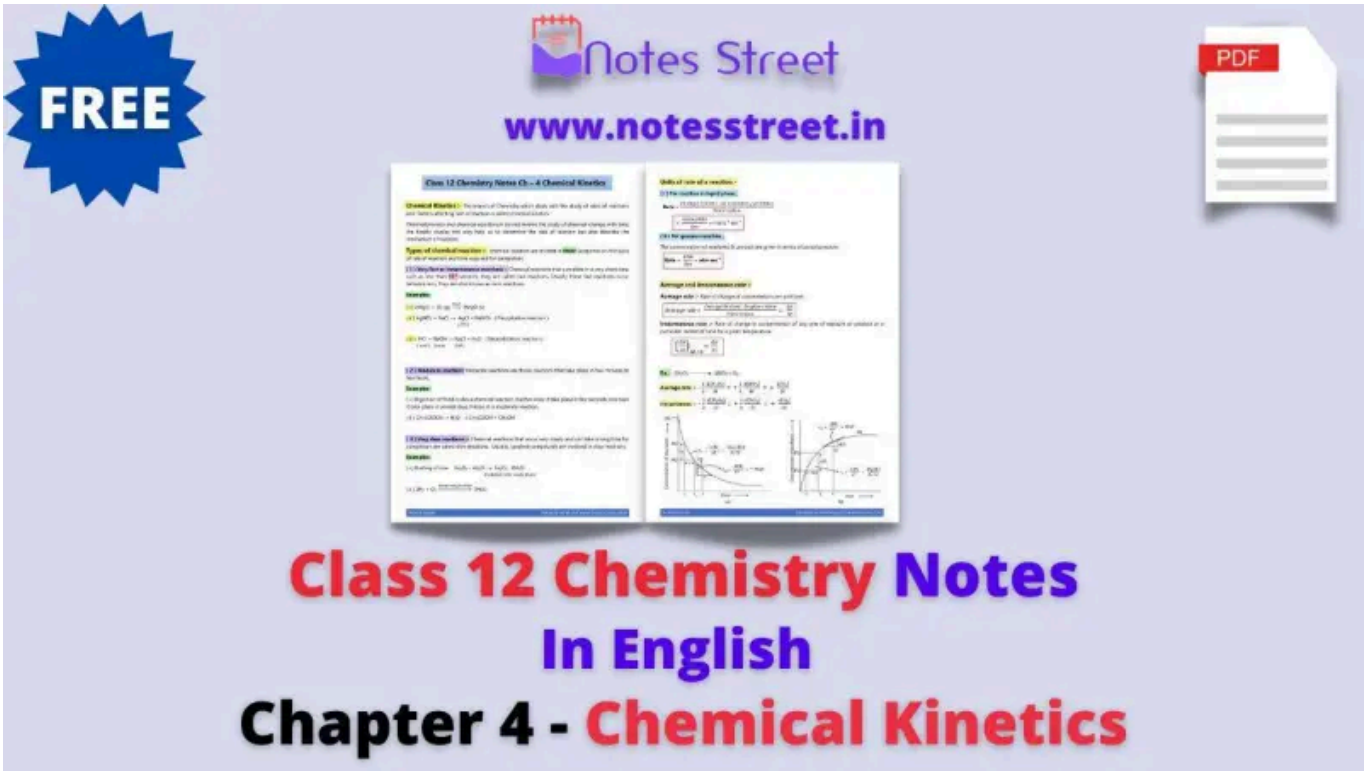


chemical kinetics class 12,class 12 chemical kinetics,class 12 chemistry chapter 4,chemical kinetics class 12 one shot,chemical kinetics,chemistry class 12 chemical kinetics,class 12 chemistry chemical kinetics,chemical kinetics class 12 chemistry,chemical kinetics in class 12 chemistry,class 12 chemistry chapter 4 chemical kinetics handwritten notes,chemical kinetics handwritten notes class 12 chemistry chapter 4,chemical kinetics 12 chemistry

Class 12 Chemistry Notes

Class 12 Chemistry Notes Chapter 4 Chemical Kinetics is an important topic in physical chemistry and basically helps the students to understand the various aspects of a chemical reaction. More specifically, the term ‘kinetics’ relates to the rate of change of some quantity.

Class 12 Chemistry Notes Ch – 4 Chemical Kinetics

Chemical Kinetics :- The branch of Chemistry which deals with the study of rates of reactions and factors affecting rate of reaction is called chemical kinetics.

Thermodynamics and chemical equilibrium do not involve the study of chemical change with time, the Kinetic studies not only help us to determine the rate of reaction but also describe the mechanism of reaction.

Types of chemical reaction :- chemical reaction are divided in **three** categories on the basis of rate of reaction and time required for completion.

(1) Very fast or instantaneous reactions :- Chemical reactions that complete in a very short time, such as less than 10^{-6} seconds, they are called fast reactions. Usually these fast reactions occur between ions, they are also known as ionic reactions.

Examples:

(i) $2Mg(s) + O_2(g) \xrightarrow{heat} 2MgO(s)$

(ii) $AgNO_3 + NaCl \rightarrow \underset{(PPt.)}{AgCl} + NaNO_3$ (Precipitation reaction)

(iii) $\underset{(acid)}{HCl} + \underset{(base)}{NaOH} \rightarrow \underset{(Salt)}{NaCl} + H_2O$ (Neutralization reaction)

(2) Moderate reaction: Moderate reactions are those reactions that take place in few minutes to few hours.

Examples:

(i) Digestion of food is also a chemical reaction. Neither does it take place in few seconds, nor does it take place in several days. Hence, it is moderate reaction.

(ii) $CH_3COOCH_3 + H_2O \rightarrow CH_3COOH + CH_3OH$

(3) Very slow reactions :- Chemical reactions that occur very slowly and can take a long time for completion are called slow reactions. Usually, covalent compounds are involved in slow reactions.

Examples:

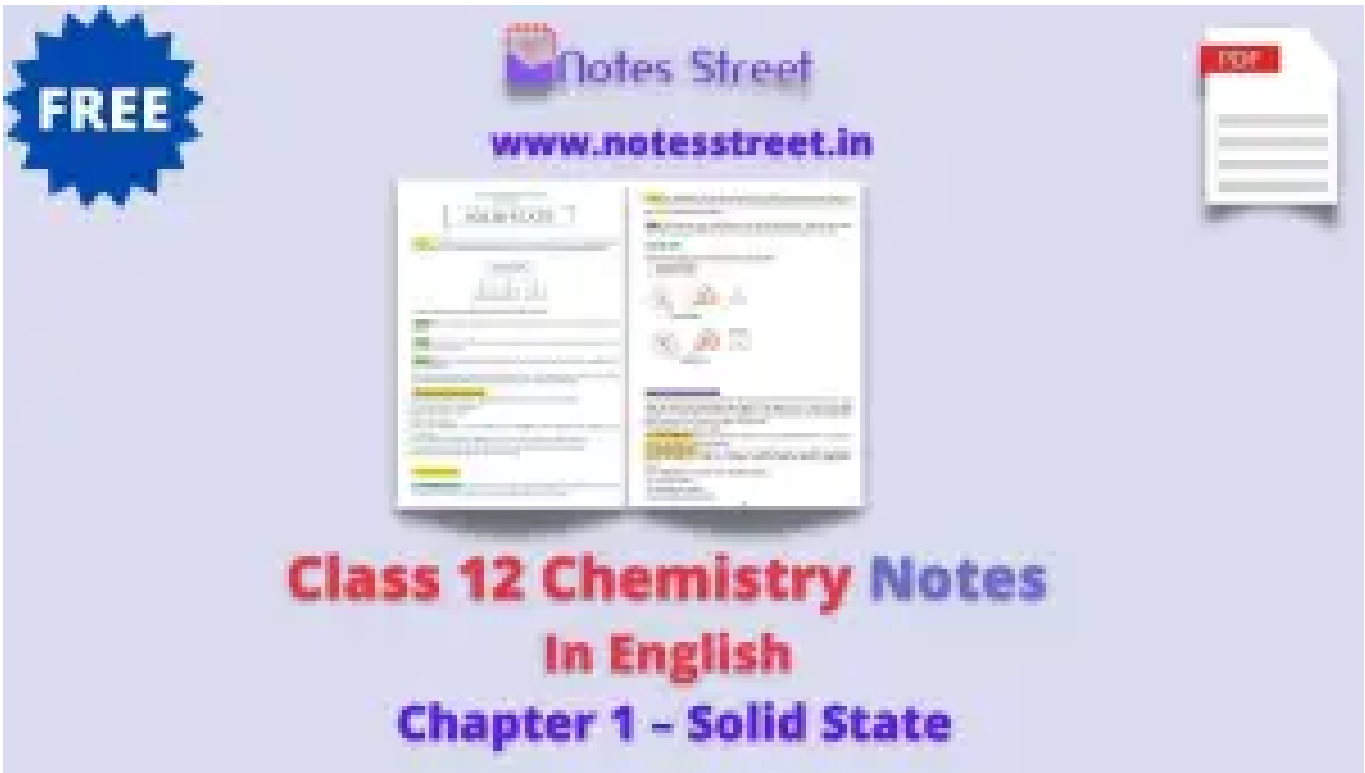
(i) Rusting of iron: $Fe_2O_3 + xH_2O \rightarrow \underset{\text{Hydrated ferric oxide (Rust)}}{Fe_2O_3 . XH_2O}$

(ii) $2H_2 + O_2 \xrightarrow{\text{Room temperature}} 2H_2O$

notesstreet.in

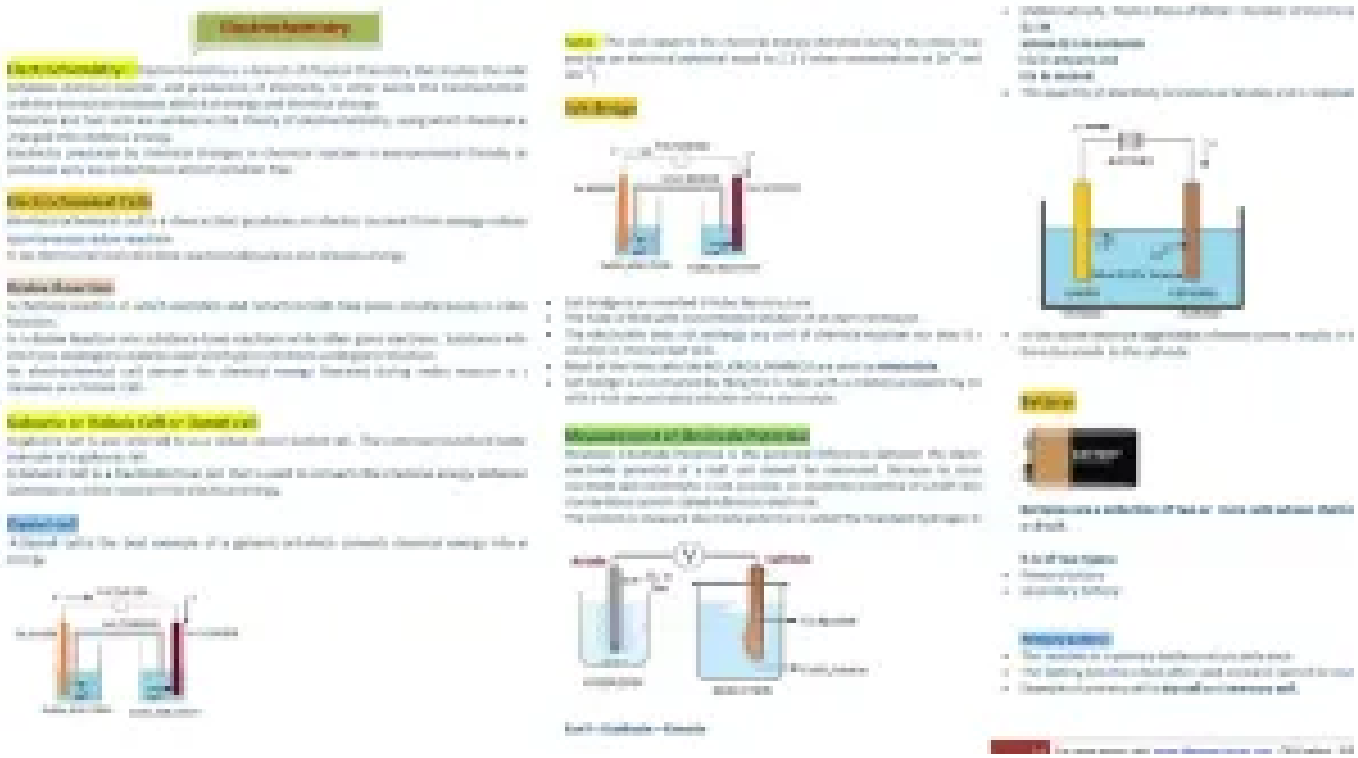
PLEASE SHARE THIS

YOU MIGHT ALSO LIKE



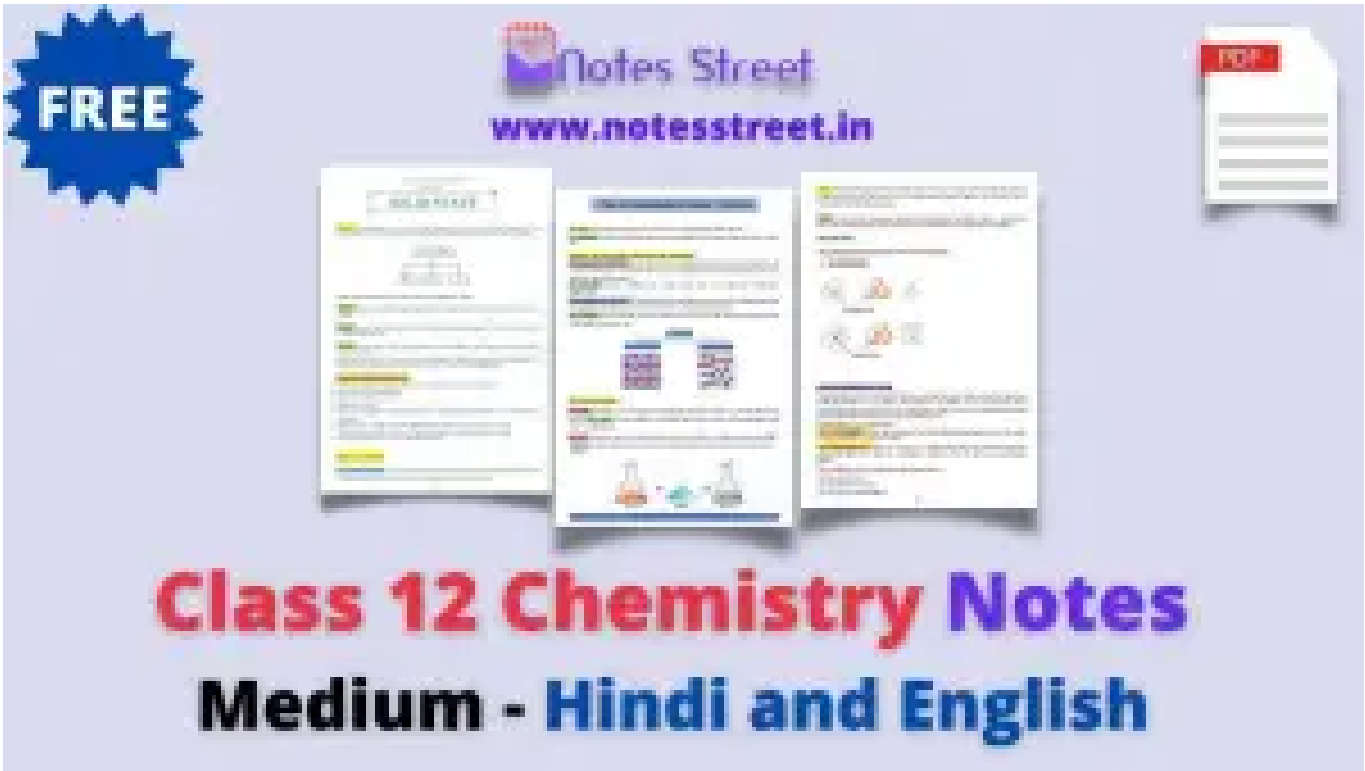
Class 12 Chemistry Notes Chapter 1 Solid State

June 21, 2020



Class 12 Chemistry Notes Chapter 3 Electrochemistry

September 16, 2021



NCERT Class 12 Chemistry Notes

June 3, 2022